Fiscal Year:	FY 2010	**Task Last Updated:**	FY 05/29/2010
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Project Title:	Molecular Basis for the Role of the Vacuole in Shoot Gravitropism		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	50011-0001	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
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No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AK78G		
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Task Description:	Plant perception and response to gravity is critical for appropriate growth, development and environmental responses. Shoot gravitropism involves the perception of gravity by sedimentation of starch-containing amyloplasts in gravity-sensing cells, followed by transmission of the signal to additional cell types to promote directional growth. Vesicle trafficking to the vacuole is essential for the perception of the gravity signal, but the molecular basis for this requirement is unknown. Our long term goal is to determine the mechanism by which gravity is perceived by shoots and the gravity signal transduced to other cell types to produce a response. The objective of this application is to determine the function of vacuolar trafficking proteins in shoot gravitropism. We hypothesize that appropriate physical properties of the vacuole are essential for amyloplast sedimentation, and therefore for gravity perception. We will test this hypothesis by measuring the elasticity of wild type and mutant vacuoles using atomic force microscopy and by		

	comparing the lipid constituents of the vacuolar membranes. We will also identify additional proteins involved in vacuolar trafficking by a combination of in vitro membrane fusion assays and in vivo immunoprecipitations and test their function in gravitropism using genetic approaches. This research will provide a molecular explanation for the observation that shoot gravitropism is specifically defective in vacuolar trafficking mutants and allow us to make testable predictions about the mechanistic basis for their agravitropic phenotype. It will also provide a foundation for the generation of transgenic plants with altered gravity response thresholds.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The primary impact of our research will be to increase understanding of the mechanisms by which plants respond to gravity. In addition to being important for the growth of plants in the altered gravitational conditions likely to be encountered during space flight, this research will also contribute to our knowledge of plant growth, development and response to environmental cues. The gravitational response interacts extensively with other developmental, hormonal and environmental response pathways, leading to morphological and anatomical changes to the plant structure. Understanding of the molecular mechanism of gravitropism is therefore vital for the development of strategies to improve plant growth and yield under varying conditions.
Task Progress:	Personnel A team of researchers has been successfully established to complete this research project. The primary researcher working on this project to date has been Dr Tony Contento, a post-doctoral Assistant Scientist, who has been developing the techniques for measurement of vacuole elasticity. Two rotation students, Eric Ottesen and Rahul Roy, have participated in the screening for SYP51 and SYP52 mutants; Rahul Roy is continuing in the lab and will continue this project for his PhD thesis research. A third PhD student, Sang-Jin Kim, has participated in setting up the in vitro vesicle fusion assays. An undergraduate student, Allison Ruby, has assisted in the research under the supervision of Dr Contento. Research progress There are two major objectives of this research: Objective #1. Analysis of vacuolar properties in the agravitropic vti11 vacuolar trafficking mutant. Objective #2. Identification of additional vacuolar trafficking components that are required for shoot gravitropism, and characterization of their molecular function. Progress toward Objective #1. Substantial progress has been made towards this aim, primarily in developing methods for measuring the elasticity of vacuolar membranes. Protocols have been optimized for the isolation of vacuoles from wild-type and gravitropism-defective vti11 mutants, and intact vacuoles from both genotypes that are suitable for elasticity measurements can now consistently be purified. Vacuoles from wild-type plants have been used in atomic force microscopy to measure the elasticity/flexibility of the vacuolar membrane. This has required extensive optimization and troubleshooting, as to our knowledge measurements of this type have not previously been performed using plant cells. We have now reached the point where consistent and reproducible results can be obtained using wild-type vacuoles, and we are beginning experiments to compare these results with vti11 mutant vacuoles to identify changes in the physical properties of the vacuolar membrane that may correlate with loss of gravitropism. Progress toward Objective #2. (a) Identification of mutants in potential shoot gravitropism genes. Our working hypothesis is that the vacuolar trafficking proteins VTI11 and SYP22, both of which are required for gravitropism, are part of a SNARE complex that delivers membrane and cargo to the vacuole. Based in previous immunoprecipitation results, we predict that the additional trafficking proteins SYP51 and/or SYP52 may be present in the same complex and therefore may also be required for shoot gravitropism. To test this hypothesis, Arabidopsis mutants with T-DNA insertions in the SYP51 and SYP52 genes have been identified by genomic PCR using primers specific to the SYP51 and SYP52 genes and to the TDNA borders. Genotyping of a segregating population of the mutants (two different alleles for each gene) has identified homozygous individuals for each SYP52 allele, which are in the process of being analyzed by RT-PCR for loss of gene expression. No homozygous mutants for SYP51 have yet been identified, raising the possibility that this is an essential gene, although further analysis is needed to confirm this. (b) Analysis of membrane fusion activity of the potential vesicle fusion proteins To biochemically analyze the function of the putative vesicle fusion proteins VTI11, SYP22, SYP51 and SYP52, in vitro liposome fusion assays will be performed using recombinant proteins to determine whether these proteins are able to catalyze membrane fusion as predicted. Constructs have been generated in E. coli expression vectors for each protein, fused to a 6xHis-tag to allow protein purification. Several of these proteins have been successfully expressed in large quantities in E. coli and affinity purified to remove unwanted contaminants. These purified proteins will be reconstituted into artificial proteoliposomes and assayed for their ability to fuse vesicles.
Bibliography Type:	Description: (Last Updated: 09/18/2019)
Articles in Peer-reviewed Journals	Zouhar J, Rojo E, Bassham DC. "Retrograde transport from the prevacuolar compartment to the trans-Golgi network." Plant Science 2010 Feb;178(2):90-3. http://dx.doi.org/10.1016/j.plantsci.2009.10.010 , Feb-2010